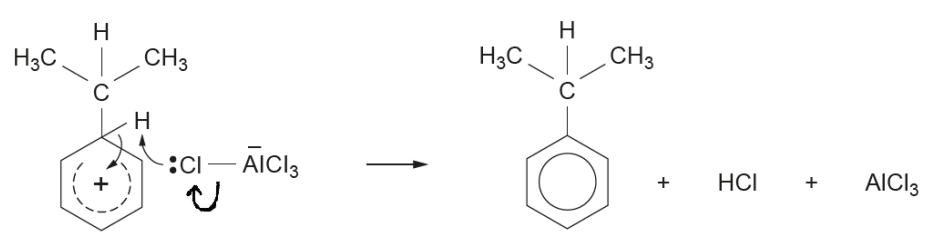
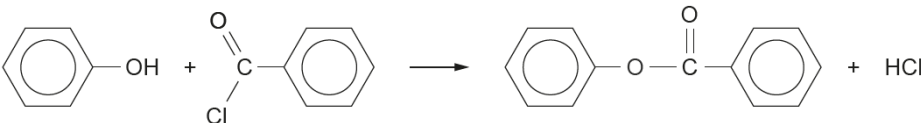


Question				Marking details	Marks available										
					AO1	AO2	AO3	Total	Maths	Prac					
9	(a)	(i)		(concentrated) nitric acid and (concentrated) sulfuric acid	1			1							
		(ii)		tin/iron and (concentrated) hydrochloric acid	1			1							
		(iii)		separation problems – the boiling temperature of the three isomers are too close together accept other sensible answers			1	1							
		(iv)		<table border="1"><tr><td>Reagent</td><td>FeCl₃</td><td>NaHCO₃</td></tr><tr><td>Observation</td><td>purple colour</td><td>no change</td></tr></table>	Reagent	FeCl ₃	NaHCO ₃	Observation	purple colour	no change	1	1		2	
Reagent	FeCl ₃	NaHCO ₃													
Observation	purple colour	no change													
	(b)			→ 5C + 2CO + N ₂ + 3H ₂ O		1		1							
	(c)			 award (1) for curly arrows – must have arrow into benzene ring and one other award (1) for all three products	1	1		2							

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)	(i)		$n(\text{benzene}) = \frac{234 \times 1000}{78} = 3000 \text{ (1)}$ $n(\text{phenol}) \text{ at } 86\% \text{ yield} = \frac{3000 \times 86}{100} = 2580$ $\text{mass of phenol} = \frac{2580 \times 94}{1000} = 243 \text{ kg (1)}$		1 1		2	1	
		(ii)		a species with an unpaired electron	1			1		
		(iii)		award (1) for any radical e.g. •CH ₃ •Cl •CH ₂ Cl		1		1		1
	(e)			award (1) for either of following solution remains yellow / orange no more white precipitate is formed			1	1		
	(f)	(i)				1		1		
		(ii)		CH ₃ COCl will react (preferentially) with the <u>NaOH / water</u>			1	1		1
		(iii)		pyridine acts as a base / removes H ⁺ (1) as its nitrogen atom has a lone pair (of electrons) (1)			2	2		
				Question 9 total	5	7	5	17	1	4